
CSE 404 : Early Dropout Risk Prediction for CSE Students at BUBT Using Institutional Learning Management Data and Explainable ML

Presented By

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June 6, 2026

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1. Introduction

Student dropout in higher education is a serious problem that negatively affects students, institutions, and national development. In Bangladesh, private universities face significant retention challenges due to financial pressure, academic difficulty in technically demanding programs, and the absence of any early warning infrastructure. Despite the scale of this problem, no institution-specific, data-driven dropout prediction system currently exists for Bangladeshi CSE programs.

Most existing dropout prediction research uses Western university datasets that do not reflect the semester structure, grading norms, course difficulty patterns, or socioeconomic conditions of Bangladeshi private universities. Models built on such datasets cannot be meaningfully applied or validated in the BUBT context.

This project proposes building a machine learning based early warning system using real institutional data from BUBT's CSE department. The system will track how each student's dropout risk changes semester by semester, identify the earliest point at which risk can be reliably detected, and use SHAP based explanations to help academic advisors understand why a student is flagged as at risk. This combination of local institutional data, temporal modeling, and explainability has not been done in any prior Bangladeshi higher education study.

2. Problem Statement

Academic monitoring at BUBT is currently reactive. Student failure is only recognized after it has already happened, often after a student has failed multiple courses or stopped attending entirely. At that point, meaningful intervention is usually too late.

Three specific problems motivate this research:

Problem 1 — No early identification system exists. There is no mechanism at BUBT that analyzes student academic records over time to flag rising dropout risk before failure becomes irreversible.

Problem 2 — No locally validated model exists. Existing dropout prediction studies in Bangladesh use small survey samples or foreign datasets. No study has used real multi-semester course level institutional records from a Bangladeshi CSE program.

Problem 3 — Black box predictions are not useful for advisors. Even an accurate model is not helpful if advisors cannot understand why a student is flagged. Explainable predictions are essential for real intervention decisions.

This project addresses all three problems together within a single framework.

3. Objectives

- To collect and analyze real institutional academic records of CSE students at BUBT including semester-wise CGPA progression, course failure sequences, and attendance patterns
- To incorporate available LMS activity data such as login frequency, submission behavior, and resource access as supplementary features
- To develop and compare machine learning models for dropout risk prediction across multiple semester windows
- To identify the earliest semester at which reliable dropout prediction is achievable for BUBT CSE students
- To analyze whether failure in gateway courses in early semesters predicts later dropout
- To apply SHAP for model explainability and generate per-student advisor-facing explanations
- To build a lightweight Early Warning System dashboard for academic advisor use

4. Related Work

Machine learning has been widely used to predict student dropout in higher education [1], [2]. Studies have consistently found that features like attendance, CGPA, grades, and LMS activity are the most important predictors of dropout risk [1], [2], [4]. Ensemble methods such as Random Forest and XGBoost have been shown to outperform single classifiers like Logistic Regression and Decision Tree on tabular academic datasets [1], [2]. Research has also demonstrated that analyzing students' digital traces and LMS log data can significantly improve early dropout prediction before failure becomes irreversible [1], [4]. Additionally, clustering techniques applied to student behavioral data have been found to enhance prediction model performance, which motivates the feature engineering approach used in this project [4]. SHAP has become the standard method for explaining machine learning predictions in educational contexts [3], and applying explainability techniques to dropout models enables personalized intervention strategies that academic advisors can directly act upon [3].

In the Bangladeshi context, no existing study has applied temporal dropout modeling with SHAP

explainability to real institutional CSE data from a Bangladeshi private university, which is the gap this project directly addresses.

5. Methodology

5.1 Data Collection

Data will be collected from BUBT institutional records after formal ethics approval from the university research committee. All records will be fully anonymized before analysis, replacing student names and ID numbers with randomized identifiers. The target cohort is CSE students admitted between 2018 and 2022, which provides complete multi-semester academic histories with observable dropout outcomes.

The following data will be collected:

- **CGPA progression** — cumulative GPA after each semester and semester-over-semester change
- **Course level grades** — individual grades for all enrolled courses, especially high failure gateway courses including Calculus, Discrete Mathematics, Data Structures, and Algorithms
- **Course failure and retake history** — number of failed courses per semester and retake patterns
- **Attendance records** — per course attendance percentage, consecutive absence streaks, and semester wise attendance change
- **Enrollment behavior** — credit load per semester, gap semesters, and leave of absence records
- **LMS activity logs** (if accessible) — weekly login frequency, late versus on-time submission ratio, and resource access patterns
- **Withdrawal records** — official withdrawal and non re-enrollment history for dropout label construction

5.2 Dropout Label Definition

A student will be labeled as a dropout if they meet any of the following conditions:

- Officially withdrew from the program
- Did not re-enroll after an unapproved gap semester
- CGPA dropped below the minimum continuation threshold with no appeal on record
- Absent from all courses for a full semester without approved leave

This definition will be validated against registrar withdrawal records.

5.3 Feature Engineering

The following additional features will be constructed from raw data:

- **CGPA delta** — semester-over-semester change in CGPA to capture decline trajectory
- **Attendance drop indicator** — flag for any semester where attendance falls more than 15% from the previous semester
- **Gateway course failure flag** — indicator for failure in high-risk courses during semesters 1 to 3
- **Cumulative failure load** — total number of failed courses up to each semester
- **Credit load ratio** — enrolled credits as a proportion of the standard full-load credits
- **Submission lateness ratio** — proportion of late submissions per semester where LMS data is available

5.4 Model Development

Three machine learning models will be trained and compared:

- **Logistic Regression** — simple interpretable baseline
- **Random Forest** — ensemble method effective on non-linear academic data
- **XGBoost** — gradient boosting with consistently strong performance on tabular datasets

Models will be trained using semester 1 data only, then semesters 1 to 2, then semesters 1 to 3, to find the earliest point at which reliable prediction is possible. SMOTE will be applied to handle class imbalance between dropout and non-dropout students. Stratified 5-fold cross-validation will be used for all model comparisons.

5.5 Explainability Using SHAP

SHAP will be applied to the best performing model to produce:

- **Global feature importance** — which features most strongly predict dropout across the full cohort
- **Temporal feature importance** — how feature importance changes across different semester windows
- **Local student level explanations** — individual risk breakdowns showing which specific factors contributed to each student's risk score, designed for direct use by academic advisors

All SHAP findings will be interpreted in the context of BUBT's specific CSE curriculum.

5.6 Early Warning System Dashboard

A lightweight Flask based dashboard will be developed for academic advisors. It will display:

- Per student risk score and semester wise risk trajectory
- Top SHAP identified risk factors for each student
- Cohort level risk distribution across semesters

- Filters by CGPA range, semester, and attendance status

6. Tools and Technologies

Category	Tool	Purpose
Language	Python 3.10+	Core development
Data Processing	Pandas, NumPy	Preprocessing and feature engineering
Machine Learning	Scikit-learn, XGBoost	Model training and evaluation
Imbalance Handling	imbalanced-learn (SMOTE)	Class imbalance correction
Explainability	SHAP	Feature importance and local explanations
Visualization	Matplotlib, Seaborn	Data exploration and SHAP plots

7. Project Timeline

Phase	Duration	Activities
Phase 1 — Preparation	Weeks 1–2	Ethics approval, data request to registrar, literature review
Phase 2 — Data	Weeks 3–5	Data collection, anonymization, cleaning, preprocessing, feature engineering
Phase 3 — Modeling	Weeks 6–9	Model development, temporal evaluation, SMOTE, cross-validation, comparison
Phase 4 — Explainability	Weeks 10–11	SHAP analysis, temporal importance, per-student explanations
Phase 5 — Delivery	Weeks 12–16	Dashboard development, paper writing, final report, submission preparation

8. Expected Outcomes

- A validated dropout risk prediction model trained on real BUBT CSE institutional data
- Identification of the earliest semester at which reliable prediction is achievable
- SHAP based insights into which academic features drive dropout risk specifically within BUBT's CSE program
- A course failure sequence analysis identifying whether early gateway course failure predicts later dropout
- A working Early Warning System dashboard usable by BUBT academic advisors
- A research paper suitable for submission to a relevant educational data mining or AI conference

References

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